g-Rel-READER: A Dataset for Relevance and Reading Evaluation through Advanced Data from Eye-tracking and EEG Recordings

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Abstract

Information relevance judgment is a fundamental cognitive decision performed by users during information seeking and reading tasks. Inferring relevance decisions from neurophysiological signals, particularly through eye-tracking and EEG recordings, opens possibilities for real-time relevance detection. To support this line of research, we present g-Rel-READER, a unique dataset comprising eye-tracking and EEG recordings collected during reading and relevance judgment of short news stories in a question-answering task. The dataset provides screen coordinates for each word, enabling temporal and spatial alignment between eye-tracking data, EEG signals, and the specific words being read. Question-relevant words are explicitly marked, facilitating analysis of relevance judgment processes.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

Keywords

Eye-tracking, EEG, Reading, Relevance judgement

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1 Introduction

Information relevance judgment is a fundamental cognitive decision performed by users during information search and retrieval. Understanding the timing and mechanisms of these judgments is crucial for characterizing user behavior and enhancing IR system design. Inferring relevance decisions from neurophysiological signals, particularly through eye-tracking and EEG recordings, becomes an intriguing possibility. Therefore, it is not surprising that recent research on relevance assessment has increasingly utilized these neurophysiological methods to provide deeper insights into the cognitive processes underlying relevance judgments [1, 5, 12, 14, 15] with promising potential for real-time relevance inference [11]. To support this line of research, we present g-Rel-READER, a unique dataset comprising eye-tracking and EEG recordings collected during reading and relevance judgment of short news stories in a question-answering task. The dataset provides screen coordinates for each word, enabling temporal and spatial alignment between eye-tracking data, EEG signals, and the specific words being read. Question-relevant words are explicitly marked, facilitating analysis of relevance judgment processes. While publicly available datasets with eye-tracking and EEG data for reading tasks exist (e.g., [10]), we are unaware of any that include relevance judgments. Our dataset provides this unique combination of signals during relevance assessment.

2 Dataset Introduction

2.1 Dataset Title

g-Rel-READER, a dataset for relevance and reading evaluation through advanced data from eye-tracking and EEG recordings.

2.2 Data Description

We share timestamped text stimulus information with relevance judgments, eye fixations, raw eye gaze, including pupil diameter, 14-channels EEG, and screen coordinates for all words from the text stimuli (news stories). All data files are in comma-separated format. The detailed data descriptions are on Open Science Foundation website: https://osf.io/3bvtx/

2.3 Related publications

Portions of these data have been analyzed and published in the following papers [3, 6–9]. Our results were partially replicated by [2].

2.4 Funding

This study was funded, in part, by the Google Faculty Research Award for the project titled: "Implicit Detection of Relevance Decisions & Affect in Web Search."

3 Methods

The data was collected in an Institutional Review Board (IRB)-approved lab-based experiment at Rutgers University.

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3.1 Participants

The study recruited twenty-four participants (15 males, 9 females; 66.7% were 18-22 years old, 16.7% 23-27, and the remaining 16.7% older than 32) from undergraduate and graduate programs at Rutgers University. All participants were native English speakers with normal or corrected-to-normal vision (20/20). The participants were asked to find information in short news story texts in response to questions. Each participant was compensated with USD 25 upon completion of the experimental session, which lasted approximately 75 minutes.

3.2 Apparatus

The experiment was conducted in a controlled laboratory environment equipped with a Windows 7 PC. Eye movement data was collected using a 17" Tobii T-60 eye-tracker operating at 60 Hz (60 samples per second), with data recorded through Tobii Studio software, which also captured keyboard responses. EEG signals were collected using an Emotiv EPOC wireless headset (Figure 1) operating at 2,048 samples per second (internally downsampled to 128 samples/s) across 14 channels, positioned according to the international 10/20 EEG format. EEG signals were recorded using Emotiv TestBench software v1.5.1 from the Emotiv Development Kit Research software package v1.0.0.3. While the Emotiv EPOC device wirelessly captures EEG signals, the acquired signals represent a combination of electrical activity from multiple sources: brain activity (EEG), eye movements (EOG), and other physiological signals including facial muscle activity, particularly at prefrontal electrode locations (AF3 and AF4). Participants were seated 65-75 cm from the monitor under standard fluorescent ceiling lights. Text stimuli were presented using custom software that controlled timing according to the experimental design. The text display was optimized for eye-tracker accuracy, with lines spanning approximately 0.98 degrees of visual angle (32 pixels) using 19pt Verdana font, and confined to a screen region of 1020x900 pixels. This display region was intentionally inset at least 100 pixels from all screen edges to compensate for reduced eye-tracker accuracy near screen boundaries.

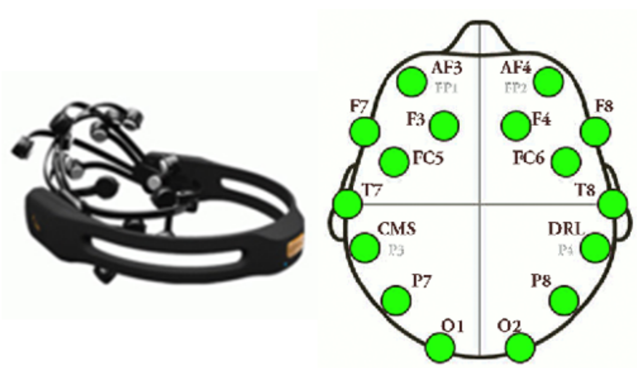


Figure 1: Emotiv EEG device and 10/20 EEG electrode positioning

3.3 Experimental Procedure

The experiment had within-subject design, where each participant performed the following two types of tasks: (a) target word search (WS task) and (b) a simple question/ answer information search (IS task). The experiment started with a WS and IS training tasks. WS and IS task blocks were presented in balanced order.

The IS task is a simulated search task with human relevance judgments. The question that was shown to participants before they saw news stories, can be considered a simulated query with three news story texts as the top three search results. Overall, in the IS task block there were 21 questions followed by three short news stories each (the trials). In our experiment, this block lasted a mean of 38 minutes (*SD* = 2.5). Figure 2 shows the stimulus sequence and timing for a single IS QA instance. The text documents were selected to cover a diversity of news topics. The experiment was presented using a fully automated process with controlled timing except for the news text display duration, which ended when the participant pressed a key to indicate their relevance judgment.

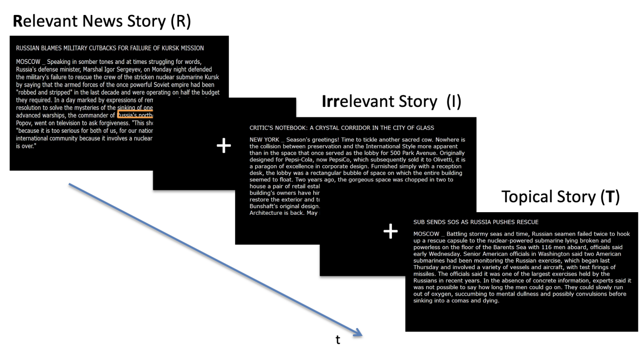


Figure 2: Sequence of one IS question with three text documents.

The IS task block started with general task instructions (30 seconds). The task goal was to find factual information in news stories that provided an answer to a question. The information target was presented as a question (8 seconds). Then a fixation cross appeared for 4 seconds to center the participant's eye gaze. Then the first news story was shown, followed by another fixation cross (4 seconds). To remind participants of the current question, it was repeated briefly (4 seconds) before the second and third document (Figure 2 "target info"). Fixation crosses were inserted between each of the news stories and the question reminders. Participants were asked to decide if the document contained an answer to the question or not and so a binary scale was used. The questions were factual in nature, for example: "Which Russian fleet was submarine Kursk part of?" News-story texts were presented for up to 20 seconds or until the participant pressed a key. This time limit was learned in pilot testing to ensure that participants could comfortably read the story and make a decision. The questions were presented in randomized order. Within a question block there were three trials; each trial was a news story text that was either:

- *Irrelevant* (I): a news-story on a topic different from the question,

- *Topical*, or partially relevant (T): a news-story on the question's topic, but not containing the question answer,
- *Relevant* (R): a news-story containing an answer to the question.

The new-stories were rotated within a question block in a pseudo-random manner using the following process. Each block contained one relevant document (R) and a combination of topical (T) or irrelevant (I) documents. Thus, within each question block there were the following three possible combinations of relevance levels: (a) RTT, (b) RTI, (c) RII. These combinations can be permuted in three, six, and three ways, respectively, yielding 12 orders: for RTT combination: RTT, TRT, TTR, for RTI: RTI, RIT, IRT, ITR, TRI, TIR, and for RII: RII, IRI, IIR. These 12 orders were used to create a sequence of 21 question blocks (nine of them were repeated twice). The sequence of 21 question blocks was randomized. We followed this procedure to avoid exposing participants to similar orders of document relevance from which they could plausibly learn patterns and try to guess relevance levels. A participant saw each document exactly once. It is important to note that participants were told that in any block some, all, or none of the stories could be relevant. This was done to enhance the ability to treat each text as involving an independent relevance judgment.

The 21 IS-task-question blocks were further divided into "*overt*" and "*covert*" blocks. In the *covert* blocks participants were instructed to make a purely mental relevance decision. The *covert* block contained 7 IS-task-question blocks. The *overt* block contained 14 question blocks (with 14*3 = 42 trials). In this block participants rated document relevance explicitly by pressing one of two keyboard buttons marked "yes" or "no." These explicit ratings of document relevance by participants are called *perceived relevance* . We call documents judged as relevant *Rp-trials* and documents judged irrelevant *Ip-trials*. In contrast to *covert* blocks, the participant action of pressing a button to confirm the relevance judgment in an *overt* block activates neural pathways associated with motor control planning and taking physical action.

Experimental procedure for WS and IS tasks is illustrated in Figure 3.

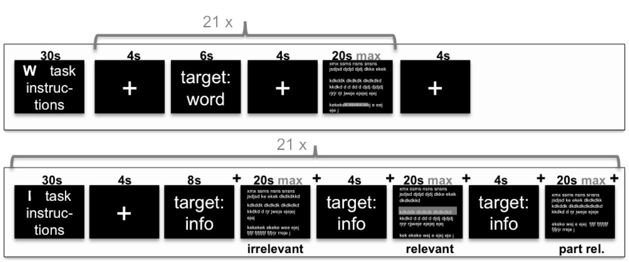
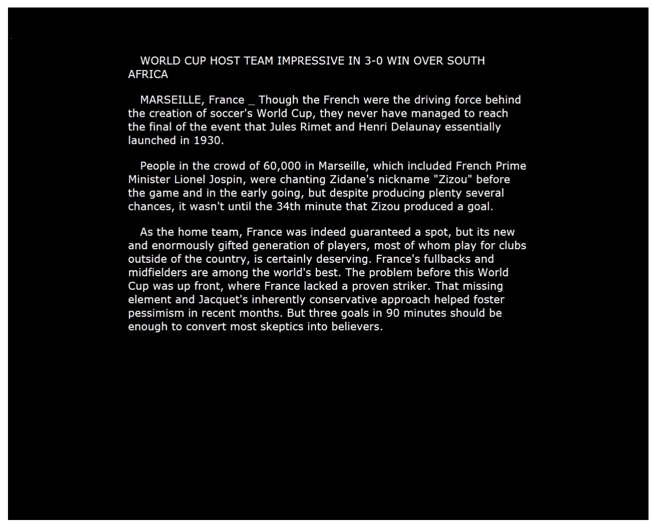


Figure 3: Flowchart of the experimental procedure.

3.4 Document Corpus

Text documents used in the study came from a large set of news stories available as the AQUAINT corpus [4]. All news stories were in English and originated from several international sources, such as Associated Press, New York Times, and Xinhua. We selected a small subset of stories (initially about 200) aiming to achieve a relatively low variation in the text length. We obtained task questions along with their relevance assessments for the documents from TREC 2005 question answering (Q&A) task [16]. We further selected 21 documents for WS task, 63 for IS task and 2 for trainings tasks and manually verified the relevance assessments for the selected document subset. We show one sample new story 4. Given the simplicity of the selected question answering fact-finding tasks, we did not need to perform inter-rater agreement assessment. These ratings constitute relevance ground truth and are referred to as *document relevance*. We processed the news stories into experimental stimuli using the following procedure. First, we created HTML documents from the new stories and added JavaScript functions and CSS tags to automate capture of AOIs for each word as it was rendered on screen. Second, we marked up relevant words, i.e., words that contained an answer to the question, in relevant documents and stored these words along with their co-ordinates in a database. Third, we took a screen shot of each HTML document without any markup of the relevant words and used the resulting images as text stimuli. The same screen resolution was used in presentation.



WORLD CUP HOST TEAM IMPRESSIVE IN 3-0 WIN OVER SOUTH AFRICA

MARSEILLE, France _ Though the French were the driving force behind the creation of soccer's World Cup, they never have managed to reach the final of the event that Jules Rimet and Henri Delaunay essentially launched in 1930.

People in the crowd of 60,000 in Marseille, which included French Prime Minister Lionel Jospin, were chanting Zidane's nickname "Zizou" before the game and in the early going, but despite producing plenty several chances, it wasn't until the 34th minute that Zizou produced a goal.

As the home team, France was indeed guaranteed a spot, but its new and enormously gifted generation of players, most of whom play for clubs outside of the country, is certainly deserving. France's fullbacks and midfielders are among the world's best. The problem before this World Cup was up front, where France lacked a proven striker. That missing element and Jacquet's inherently conservative approach helped foster pessimism in recent months. But three goals in 90 minutes should be enough to convert most skeptics into believers.

Figure 4: Sample news story used in this experiment

3.5 Data Preprocessing and Cleanup

Step 1: Poor-quality Eye-tracking data records (Tobii validity <4) and off-screen fixations were removed (5̃% of fixations). Minimal continuity of eye-tracking data was enforced. We removed trials with gaps longer than 1 second [13], or >20% missing data, or fewer than four fixations on a document. Remaining trials with small eye-tracking data gaps, for example as a result of eye blinks, were treated by merging pupil measurements for both eyes. If both eyes were captured the mean was used, one eye was used if there was no measurement for the other. A linear interpolation algorithm was applied to the remaining pupil data gaps.

Step 2: The relevance level (I/T/R) distribution of the remaining trials was not equal. To gain a more uniform distribution of document relevance degrees and text lengths, we removed data from "long" (> 310 words) irrelevant documents to better match the document characteristics of the remaining topical and relevant trials. The resulting trial collection had an average document length of 178 words (SD = 30) with relevance level distribution: I/T/R (19/21/19).

4 Usage Information

4.1 How to cite

Gwizdka, J., Cole, M. J. (2024). g-Rel-READER: A Dataset for Relevance and Reading Evaluation through Advanced Data from Eye-tracking and EEG Recordings. Retrieved from osf.io/3bvtx

4.2 Simple Example

Example in Figure 5 demonstrates temporal alignment of pupil dilation signal (showing both raw and smoothed measurements) with fixated words during text reading. Word fixations are marked at their corresponding time points along the pupil dilation signal.

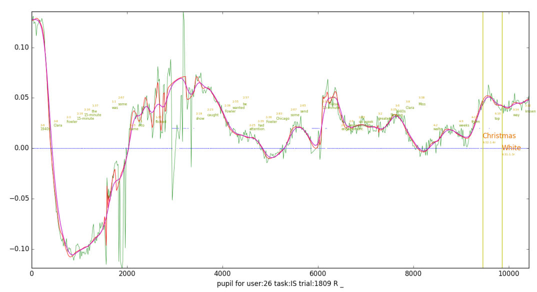


Figure 5: Pupil dilation during reading, showing both raw and smoothed signals, with corresponding fixated words (green font for regular words and orange for relevant words) indicated at their respective timestamps

4.3 License Type

This dataset is licensed under the Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0).

5 Conclusion

The g-Rel-READER dataset provides a unique contribution to the study of information relevance assessment by combining eye-tracking and EEG data with explicit relevance judgments in a question-answering context. The dataset's distinctive features include: (1) precise temporal and spatial alignment between eye movements, EEG signals, and specific words being read; (2) explicit marking of question-relevant words enabling fine-grained analysis of relevance judgment processes; and (3) carefully controlled experimental conditions with balanced presentation of relevant, topical, and irrelevant documents.

The dataset was collected through rigorous experimental procedures involving 24 participants performing two complementary tasks: target word search (WS) and question answering information search (IS). The WS provides a baseline reading behavior data. The IS task provides deeper insight into relevance assessment through both explicit ('overt') and implicit ('covert') relevance judgments across 63 news stories. The careful preprocessing and data cleanup ensure high data quality, with removal of poor-quality recordings and standardization of document characteristics. Unlike existing reading datasets such as Zuco [10], this dataset specifically incorporates relevance assessment, making it particularly valuable for researchers investigating the cognitive processes underlying information relevance judgment, as demonstrated by several published studies utilizing and replicating findings from these data.

The g-Rel-READER dataset, available under a CC BY-NC 4.0 license, offers opportunities for researchers to advance our understanding of relevance assessment processes, develop improved information retrieval systems, and explore the potential of real-time relevance inference from neurophysiological signals. Future research may use this dataset to develop more sophisticated models of user behavior during information search and to create more responsive information retrieval systems.

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